Evaluation methodology for structured light 3D based depth maps

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ABSTRACT

Structured light depth map systems are a type of 3D system where a structured light pattern is projected into the object space and an adjacent receiving camera is used to capture the image of the scene. By using the distance between the camera and the projector together with the structured pattern you can estimate the depth of objects in the scene from the camera.

It is important to be able to compare two systems to see how one compares to another. Accuracy, Resolution, and Speed are three aspects of a structured light system that are often used for performance evaluation.

It would be ideal if we could use the accuracy and resolution measurements to answer questions such as how close two cubes can be together and be resolved as two objects. Or, determine how close a person must be to the structured light system in order to determine how many fingers this person is holding up.

In order to be able to do this it is necessary to choose a target or a set of targets from which we make measurements that will enable us to quantify how well a system can resolve objects that are close to each other and how well the system can determine the shape of and object.

In this document we will go over a number of targets we evaluated and the "Cut-out Siemens Star" target that we selected as being the best choice. Using this target we evaluated the performance of two systems. The metrics we used for the evaluation were developed during this work. Using these metrics you can compare how well one system is able to resolve two objects that are close to each other and how well the system is able to discern the shape of an object.

1. INTRODUCTION

1.1 Background

Structured light depth map systems are a type of 3D system where a structured light pattern is projected into the object space and an adjacent receiving camera is used to capture the image of the scene. By using the distance between the camera and the projector together with the structured pattern you can estimate the depth of objects in the scene from the camera.

There are different types of structured patterns that can be projected into a scene. Most of the patterns types are described in [1]. Some possible patterns are binary coded, color coded, gray scale coded, pseudo-random binary dots and patterns made up of mini-patterns that form coded words. In practical applications, we may want to know how well a structured light system is able to resolve, for example: two cubes placed close to each other, two fingers someone is holding up or the shape of a small object (e.g. is the object square or round)

It is important to be able to compare two systems to see how one compares to another. There are three aspects of a structured light system that are often used to evaluate the performance of a system [1]:

Accuracy: Refers to how accurate the measured x, y and z values are with respect to the "ground truth" [2], where z is the depth coordinate and x and y are the horizontal and vertical coordinates, respectfully. The "ground truth" is the actual x, y and z location of a point in the object space. Accuracy, therefore, is the deviation of the measured value as compared to the actual ground truth values. For each 3D image captured by the structured light system one can express the errors of all measured locations in terms of mean error, standard deviation, maximum error and etc.

Resolution: Refers to the ability of a system to resolve a point in the object space. You can measure this by determining the number of reported depth pixels provided by a system, or you can make measurements that will tell you how well a system is able to resolve two adjacent points in the object space. Other methods can also be used, such as comparing the shape of a known object, such as a cube or sphere, to the shape obtained in the depth map provided by the structured light system [2].

Speed: Refers to the rate at which the system is able to capture and process a depth map image. Some systems can achieve 30fps or more. This facilitates tracking the movement of objects.

Some systems use post processing to improve the accuracy of the depth map information it measured. For example, a system can look at each depth map point it measured and compare it with the neighboring points. If the measured depth of that point is far in distance from the neighboring points, then the system could consider this measurement erroneous. In which case, it could use the neighboring measurements to correct the point it thinks is erroneous.
Such type of post processing can make it difficult to find a target that will not bias the results. For example a 3D target system made up of two small cubes close together could produce results different than that for a system with just one small cube. For the target with two cubes, the post process can cause the gap between the two cubes to merge together forming one object instead of two. Whereas the post processing may not have as much of an impact if only one cube is used.

Another issue is the need to know the "ground truth" in order to measure the accuracy, resolution and speed. One way to do this is to use accurate depth measurement equipment in order to measure the ground truth. This could be tedious and time consuming. It would be good if we could find a target that will enable comparing two systems without requiring efforts to determine the ground true.

The orientation of an object can also influence the results. For example the results can be different if you rotated a cube 45 degrees with respect to the three x, y and z axes. It therefore becomes important to consider how the orientation of an object or objects will affect the accuracy and resolution of a system. One could take a series of measurements where a cube, or a different 3D target, is rotated by a quantized amount, but the process of taking many such pictures could become overwhelming. It would be good if we could find a target that would minimize such an effort.

The goal of our research was to find a target that would compare the resolution of two systems without the need to know the ground truth and without having to assure the orientation of the target is the same for both systems.

1.2 Our approach

Our goal is to develop a set of metrics based on targets that will enable us to quantify the performance differences between two or more structured light systems. We need to be able to quantify the performance as a function of depth along the z axis and along the x and y axis. Our criteria for such a test system is to be able to use a flat test target that is perpendicular to the z axis, where this target can be moved from the systems closest operating distance to the farthest distance. The idea is to determine how well the system can resolve flat objects that reside on this plane. We can then make measurements at a specific set of distances from the system measured along the z axis. For each distance we want to use metrics that don't require knowing the ground truth and are insensitive to the orientation of objects found in the plane of the target.

We have developed a test target and methods to evaluate the ability of a structured light system to resolve objects and edges. Our objective was to make a set of measurements that would enable us to assess how well a system performs.

2. TEST TARGET

2.1 Target Design criteria

The two areas we are focusing on are the ability of a system to resolve two separate objects that are near each other and how well a system can resolve the shape of an object. Ideally we want a target that does not require knowledge

of the ground truth and does not require tedious alignment procedures. The target should also provide for a way to removed dependencies on the orientations of an object or objects.

2.2 Test target considered during the design process

During our research we tried a number of targets. The target that we ultimately selected is called the "Cut-out Siemens Star" and is shown in figure 1. We looked at other targets as well. Some of these targets are described below and shown in figures 2, 3 and 4.

2.2.1 Cut-out Siemens Star target

Figure 1 shows a target that is commonly referred to as the Siemens Star and used to measure resolution in typical 2D pictures. The Siemens Star has light gray areas and black areas as shown. The targets we propose for testing a structured light system is the same, except the black areas are cut out, which creates holes that look like pie wedges. It is important to point out that in the center there is a circle that has 10% of the full 100% radius shown in the picture. The area within this 10% circle is not cut out. This center circle area is used to attach the gray spokes together in the center.

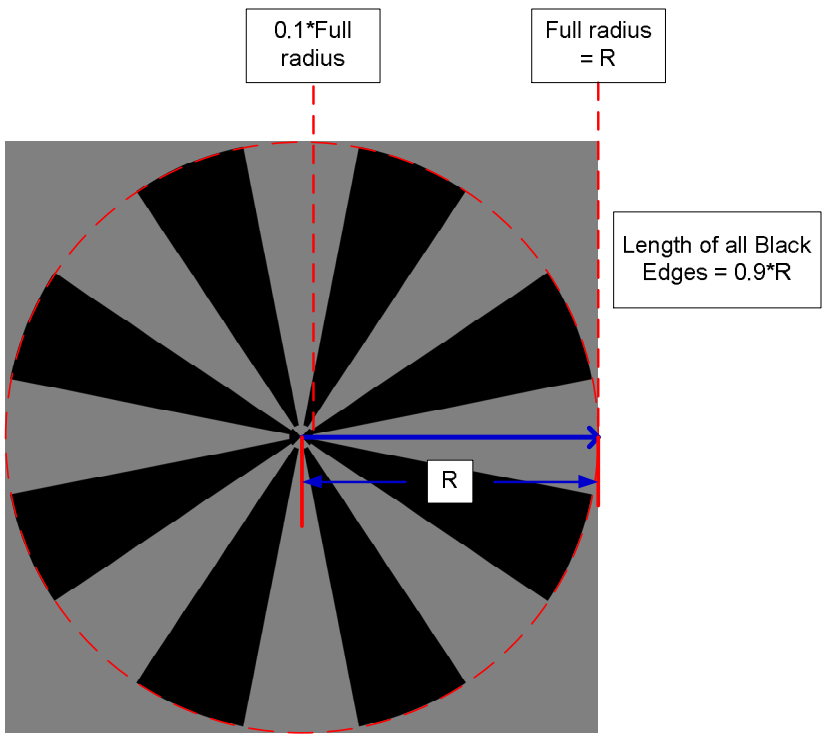


Figure 1. Siemen star

2.2.2 Cut-out bars of decreasing widths and its complement

These are two complementary targets (see Figure 2). For both, the black areas are cut out creating holes. The smallest hole shown on the top target is $1/10^{th}$ of the adjacent bar width, where all bars in the chart are of the same width. Likewise, the smallest bar in the lower target is $1/10^{th}$ of the adjacent hole, where all holes are of the same width.

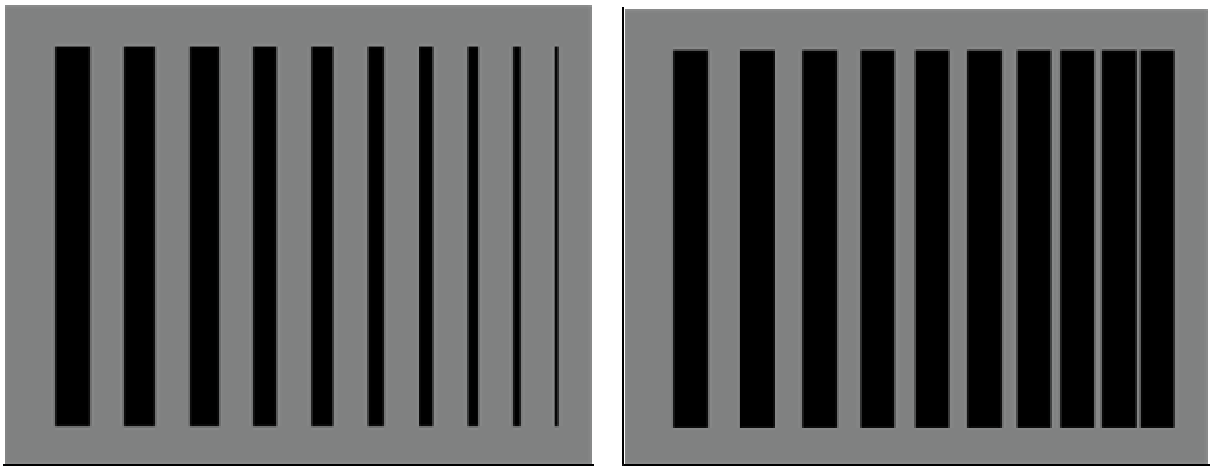
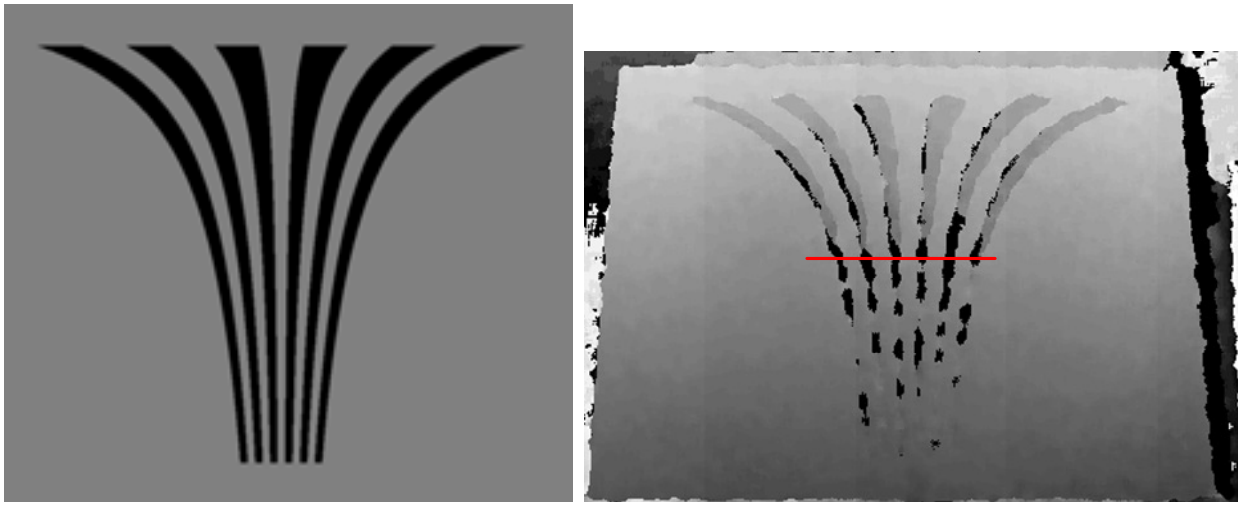


Figure 2. Cut-out bars of decreasing widths and its complement

In order to use this target to compare two structured light systems we need to find a way to assure each structured light system projects the same number of depth pixels on the target. We could do this by aligning the target so it fills the full Horizontal Field of View (HFoV) and then measure from the captured depth map picture the number of horizontal depth pixels across the target. We can do the same for another system. Using this information we can figure out where to place the target for one system relative to the other system. Once we know the same number of depth pixels cover the target between two systems we can then determine what is the smallest bar or hole each system can detect. We will then need to rotate the target by 45 and 90 degrees and make the same measurements. It is possible these three orientations will yield different results. Using these measurements we will then need to develop a merit function to determine which system performs better and by how much.

2.2.3 Cut-out Curved Lines of Variable Width

Figure 3 shows a pattern that progressively degrees from the top to the bottom. The black areas are cut out creating holes. If you draw a horizontal line across the bars you will find the holes and bars will have the same width at that location. The target uses a progressively decreasing width pattern from top to bottom. The idea behind this target is to find the horizontal line location where the system is just barely able to detect all holes and bars along that line, like that shown in figure 3(b). This figure shows the depth map pictures captured by a structured light system. The red line shows the horizontal line where the system is just barely able to detect all the holes and bars along this line. For each system we can then find the average number of depth pixels along that line that fit in the holes and the average number that fit over the bars. The next step is to rotate the target by 45 an 90 degrees and make the same measurements. If the lines are far apart from each other on the target the differences in the curved pattern may bias one system relative to the other, so it is best if the target distances are aligned to assure the lines are near the same location on the target.



(a) (b)

Figure 3. Cut-out curved lines of variable width

2.2.4 The Cut-out Checkerboard Target

Figure 4 shows a checkerboard pattern, where the center most black areas have been cut out. The idea is to create holes around the center most square. Using this chart allows us to investigate how well the structured light system under test is able to distinguish the center square as being square. It is possible under certain conditions that the detected shape looks more like a circle rather than a square.

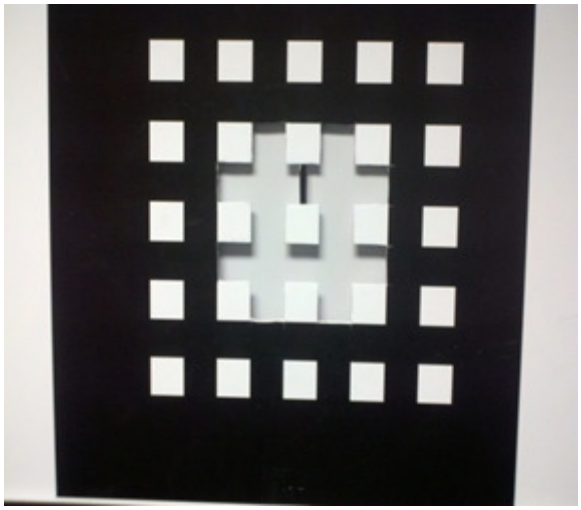


Figure 4. Cut-out checkerboard target

2.3 Selected test target: The Cut-out Siemens Star Target

Based on the design criteria, we have found that the cut-out Siemens star test target, shown in Figure 1, better fits our requirements. The star target is less influenced by the targets orientation and distance. All of the other targets require careful alignment procedures in order to accurately compare one structured light system to another.

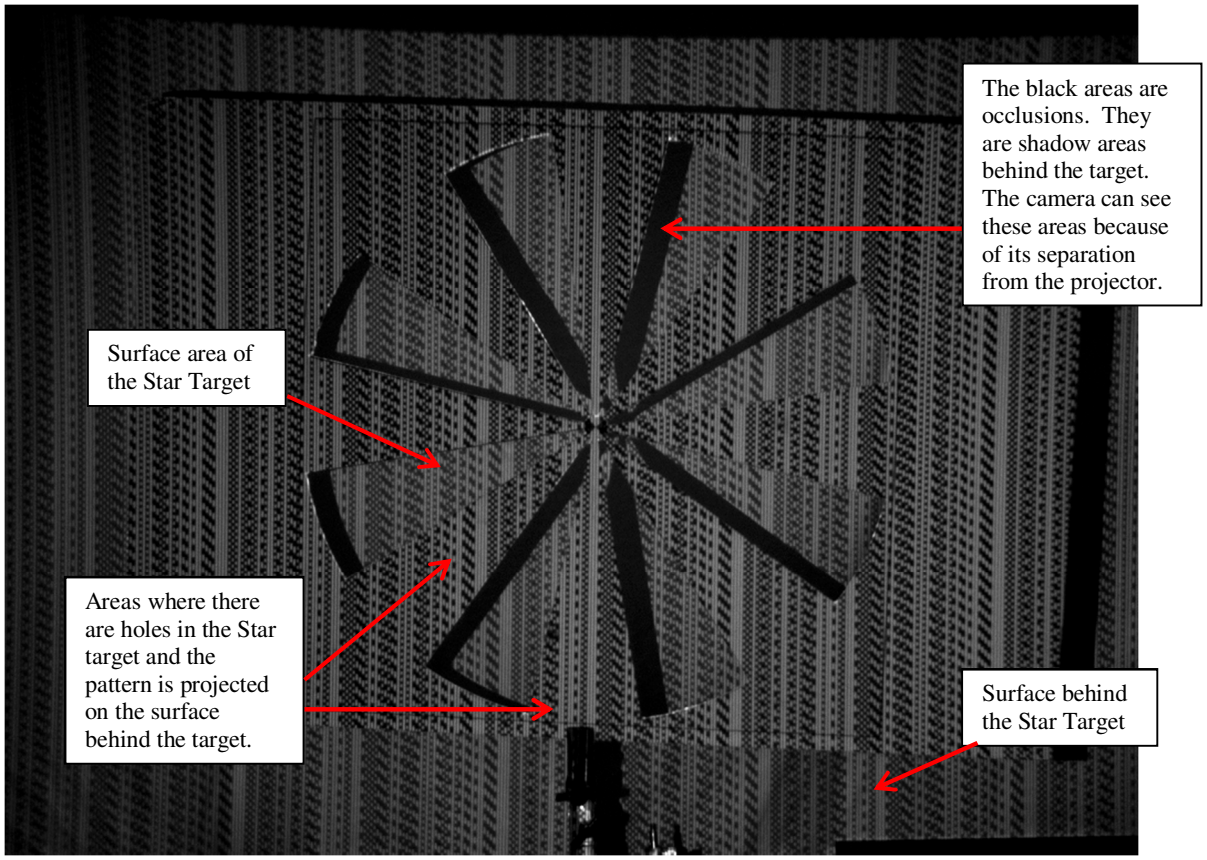
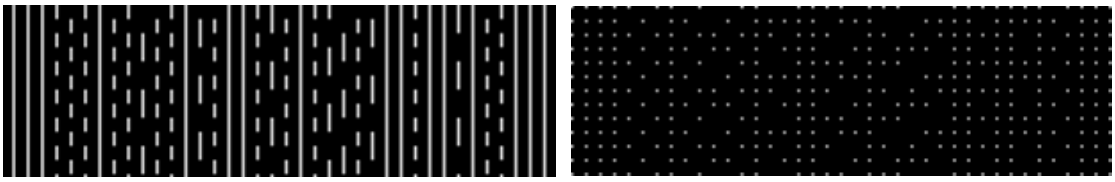


Figure 5. Image captured by the IR camera using the cut-out Siemens Star Target

Figure 5 shows the image captured by the IR camera. It shows the projected structured light pattern on the Star target and the surface behind the Star Target. The structure light pattern is based on vertical lines as shown in Figure 6a; other patterns may use dots, as shown in Figure 6b. Notice that the cutout pie wedges show the pattern on surfaces behind the target. The black areas are shadows.



(a) (b)

Figure 6. Structured-light pattern: (a) lines, and (b) dots

3. MEASUREMENTS MADE USING THE Cut-out Siemens STAR Target

3.1 Performance metric

The ideal Star target is shown in Figure 1. The image has been cropped so that the distance from the center of the picture to the horizontal or vertical edges will be the same length from the center to the end of the black areas in the picture. The length of any black edge is equal to 90% of the Full Radius, R, in depth pixels.

Let's assume:

- Each pixel in this picture represents a depth pixel
- The gray areas are depth pixels found on the targets surface

- The black areas are holes. These pixels when captured by the camera will either be in shadows, on surfaces behind the target or depth errors made by the structured light system.

With these assumptions we can compare the number of gray pixels, "Pg", laying on the target surface and on a circle with a radius of

$$Rz = Z*R, \text{ where Z is scalar from 0.1 to 1.0} \qquad (1)$$

We can then calculate the percentage

$$Y = 100*(Pg/(Pi*Rz)) \qquad (2)$$

Ideally we would expect Y to be equal to 100% for all radial distances Rz.

Figure 7 shows the results of computing Y for the target shown in Figure 1. As you can see the Y turns out to be 102 percent. This is because the target in Figure 1 has one more gray pixel than black pixels along any of the circles with radius Rz, where Rz > 0.1*R. It turns out this difference is not as significant as it may seem. This will become more evident as we describe actual test results.

Let's use the symbol X to represent the numbers on the abscissa of Figure 7.

$$X = [(Rz - 0.1*R)/(0.9*R)]*Xmax, \text{ where Xmas is the largest value for X in Figure 7} \qquad (3)$$

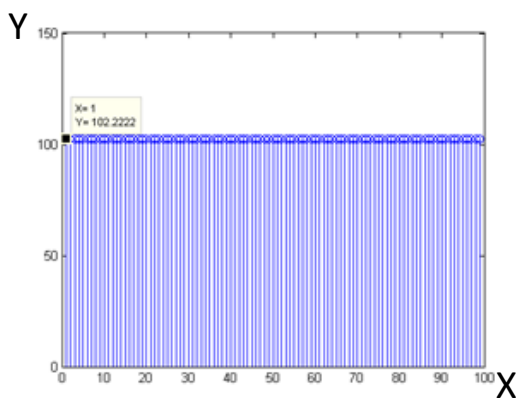


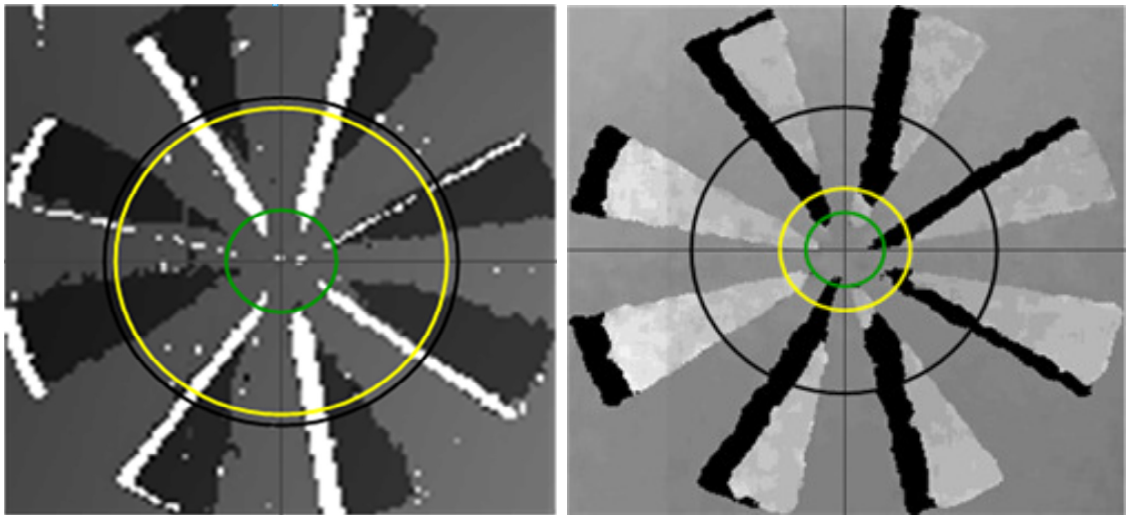
Figure 7. Evaluation results for target in Figure 1

3.2 Performance evaluation of 2 systems

The two systems we use for comparison are:

- A system that we've developed in our lab that projects a line pattern, which we will call the Q-Line System
- A third party system that projects a dot pattern, which we will call the C-Dot System

In Figure 8, cropped depth map images are shown for the Q-Line and C-Dot systems. Pictures (a) and (b) were cropped so that the horizontal and vertical edges of the pictures are about twice the radius of the circle encircling the outer edges of the pie wedge holes. By cropping the pictures in this way we can make the assumption that the center of the captured target is in the center of the picture. Granted, it is not likely to be, but taking care to align the target perpendicular to the optical axis of the camera and having the center of the star target very closely aligned with the optical axis you should get a good enough picture to make meaningful measurements. In particular, the Q-Line system depth map picture which has been cropped to 126x126 depth pixels; and the C-Dot system picture that has been cropped to 278x271 depth pixels.



(a) (b)

Figure 8. Depth Map: (a) Q-Line System, and (b) C-Dot System

The results using our evaluation program, for both the Q-Line and C-Dot Systems, are shown in Figure 9.

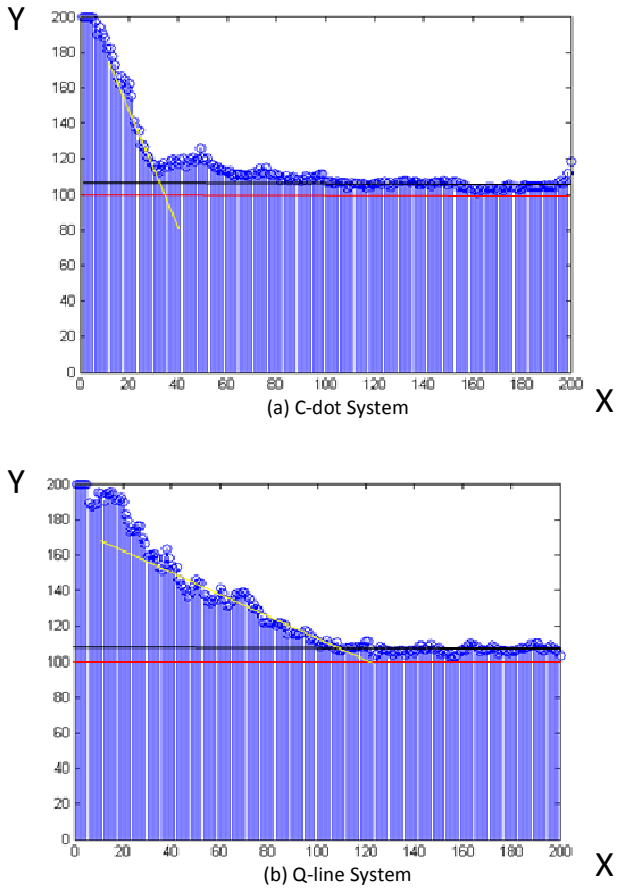


Figure 9. Performance metric: (a) C-Dot System and (b) Q-Line System,

The Q-Line image shown in figure 8b is 126x126 pixels and for the C-Dot image in figure 8a it is 278x271 pixels. Using this information we can estimate the radius R for both pictures to be:

Rk = (278+271)/2, as the radius R for the C-Dot system
Rq = 126/2, as the radius for the Q-Line system

Using these radius values we can now determine what the abscissa values mean in the graphs above by re-writing equation (3) to solve for Rz. This equation is:

Rz = [(X/Xmax)*0.9 + 0.1]*R, where Xmax is the largest X value shown on the graph. (4)

Notice the yellow line for either graph in Figure 9 approximately follows the slope of the graph and intersects the black line around the value of 30 for the C-Dot system and 110 for Q-Line system. What we want to know for each system is what the value of Rz is where the yellow line intersects the black line. It turns out

For the C-Dot system Rz at X=30 is 32
For the Q-Line system Rz at X=110 is 38

On Figure 8a and Figure 8b you will see a yellow circle. The yellow circle has a radius, Rz, of 32 for the C-Dot system and 38 for the Q-Line system.

The black lines in the graphs in Figure 9 (a) and (b) are used to identify the flat area of the data points in the respective graphs. The black line is positioned to be about the mean value of the flat area. The black circle in Figure 8(a) and (b) represents the radius at which the data values in the respective graph begin to rise above the black line. For each respective graph we find the X value for the C-Dot system is 100 and 105 for the Q-Line system. We can now calculate the Rz radius value for the black circles as follows:

For C-Dot Rz at X=100 is 75
For Q-Line Rz at X=105 is 36

We can now evaluate edge consistency by looking at the edges where Rz is longer than that for the black circle. You can observe from the images in Figure 8(a) and (b) the edge roughness is different depending on the angle of the edge. This leaves us with the possibility to evaluate the average edge consistency on all 16 edges or report the edge consistency for all 16 edges separately.

For edge consistency we will use the average over 16 edges and measure the mean, standard deviation and the min and max values about the mean as follows:

For the Q-Line system we measured

- Mean = 106.8, Standard Deviation = 1.95, Min value below the Mean = -5.2, Max value above the Mean = 4.8

For the C-Dot system we measured

- Mean =105.1 , Standard Deviation = 1.75, Min value below the Mean = -4.6, Max value above the Mean = 2.7

  [Note: In the C-Dot plot in figure 7a, the Y values between X=190 to X=200 are above the mean. We removed those values from when we calculated the Mean, Standard Deviation, Min and Max values.]

The black line in the graph represents the mean value of the data points in the flat area. The Y values for the black lines are approximately 105 for C-Dot and 108 for the Q-Line system. The numbers are an indication that the hole wedges are slightly more narrow in the depth image than the actual picture captured by the camera. Both values are close to 100 and close to each other. They are essentially the same in performance.

The space between the yellow circle and the black circle can be thought of as a special area to investigate further. For the C-Dot system, Figure 9a indicates the target surface between the yellow circle and the black circle appears wider than it is as seen by the camera. The significance or reason may warrant further investigation.

Next we want to turn our attention to the area of the graphs where Rz is less than the location where the yellow and black lines intersect.

We can make some qualitative assessments by trying to gain a concept of what Y=180 may mean in terms of detectability

- For the C-Dot system Y=180 corresponds to X=10 and Rz=19.9
    - 2*Pi*19.9 = 125 pixels around the green circle in figure 6b and c, where the green circle has radius Rz = 19.9
    - 125/16 is equal to 7.8 pixels per 16 wedges (8 for the black wedges and 8 for the gray wedges)
- For the Q-Line system Y=180 corresponds to X=20 and Rz=12.97
    - 2*Pi*12.97 = 75.2 pixels around the green circle
    - 75.2/16 is equal to 4.7 pixels per 16 wedges

These results indicate that the Q-Line system is able to detect small objects slightly better than the C-Dot system.

Now that we have made a set of observations about the Star target and carried out some measurements we can go further and define metrics for Threshold Resolution, Resolution at Y=150 and Edge Consistency.

Threshold Resolution: This is the smallest green circle, as shown in figures 8a and 8b that touches at least one depth map pixel of all 16 wedges. By measuring the radius Rz of this green circle you can calculate the number of depth pixels around the green circle by using the equation 2*Pi*Rz. The Threshold Resolution (TH) metric is then calculated as TH=2*Pi*Rz/16, which is expressed in pixels per 16 wedges.

Resolution at Y=150: This is the number of pixels per 16 wedge when the Y value, as shown in figures 9a and 9b, is equal to 150. We will use ResY-150 to represent this metric. In the same fashion one can setup a table that provides the number of pixels per 16 wedges at other values of Y, such as 170, 150 and 130. These values can be represented by ResY-170 and similarly for other values of Y.

Edge Consistency: Is defined by the mean, standard deviation, min and max values of Y, for all values of Y where X is greater than the point where the Y value begins to rise above the black line. By way of an example, this is shown in figures 9a and 9b. The black line is placed over the area where the Y values have flattened out. In the graph there is a point at which the Y values begin to raise as the X value decreases, which we will call X-Flat. Using these graphs we want to calculate the mean, standard deviation, min and max values of Y for all values of X above X-Flat.

The Threshold Resolution, TH, provides information about how close two objects can be and still be detected as one object.

Resolution ResY-150, provides a common point at which two systems can be compared. This expressed in the number of depth pixels per 16 wedges when the Y value is equal to 150. When Y is equal to 150, it means the system is reporting 50% more pixels as being on the target surface as compared to the actual number of depth pixels projected on the target by the system's projector. Ideally, we want Y to converge to 100 for a smallest value of X. ResY=150 can be thought of as an indicator of rate of how Y is converging to 100.

Edge Consistency provides a way to describe how well a system is able to discern an object's shape. For example the object could be square, but the systems depth map could show it as being round. The shape shown in the depth map could be affected by the objects orientation. By making measurements along 16 edges, as we have done above, we get numbers that provide an indication of how well a system can represent the actual shape of an object.

4. CONCLUSIONS

Our goal was to find a flat target that will allow us to compare how well a structured light system can resolve flat objects that reside in a plane perpendicular to a system's z axis. Above we have introduced the 16 wedge Star target and defined metrics for Threshold Resolution, Resolution Res-Y-150 and Edge Consistency. One can use these metrics to assess how well a system can resolve planar objects and shapes.

In our future work we will be exploring ways to assess a system's depth accuracy along the z axis.

5. REFERENCES

1. Jason Geng, "Structured-Light 3D Surface Imaging: A tutorial," Advances in Optics and Photonics **3**, 128–160 (2011)
2. Xu Zhang, Limin Zhu, Liwen Chu, "Evaluation of Coded Structured Light Methods Using Ground Truth," *2011 IEEE 5th International Conference on Cybernetics and Intelligent Systems (CIS)*